LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: HR295236
Referred BY	: Self	Sample Type	: Whole blood EDTA
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:54 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

FK Answer Genomics Package

Complete Blood Count (CBC)

Test Description	Value(s)	Unit(s)	Reference Range
RBC Parameters			
Hemoglobin *colorimetric*	**17.1**	g/dL	13.0 - 17.0
RBC Count *Electrical impedance*	5.4	10^6/µl	4.5 - 5.5
PCV *Calculated*	48.6	%	40 - 50
MCV *Calculated*	90.3	fl	83 - 101
MCH *Calculated*	31.8	pg	27 - 32
MCHC *Calculated*	**35.2**	g/dL	31.5 - 34.5
RDW (CV) * *Calculated*	11.9	%	11.6 - 14.0
RDW-SD * *Calculated*	42.2	fl	35.1 - 43.9
WBC Parameters			
TLC *Electrical impedance and microscopy*	6.1	10^3/µl	4 - 10
Differential Leucocyte Count			
Neutrophils	44.3	%	40-80
Lymphocytes	40	%	20-40
Monocytes	2.8	%	2-10
Eosinophils	**11.8**	%	1-6
Basophils	1.1	%	<2
Absolute Leukocyte Counts ***Calculated***			
Neutrophils.	2.7	10^3/µl	2 - 7
Lymphocytes.	2.44	10^3/µl	1 - 3
Monocytes.	**0.17**	10^3/µl	0.2 - 1.0
Eosinophils.	**0.72**	10^3/µl	0.02 - 0.5
Basophils.	0.07	10^3/µl	0.02 - 0.5
Platelet Parameters			
Platelet Count *Electrical impedance and microscopy*	300	10^3/µl	150 - 410
Mean Platelet Volume (MPV) * *Calculated*	9.8	fL	9.3 - 12.1
PCT * *Calculated*	0.3	%	0.17 - 0.32

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: HR295236
Referred BY	: Self	Sample Type	: Whole blood EDTA
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:54 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range
PDW * *Calculated*	16.9	fL	8.3 - 25.0
P-LCR * *Calculated*	32.8	%	18 - 50
P-LCC * *Calculated*	98	10^9/L	44 - 140
Mentzer Index * *Calculated*	16.72	%	> 13

Interpretation:
CBC provides information about red cells, white cells and platelets. Results are useful in the diagnosis of anemia, infections, leukemias, clotting disorders and many other medical conditions.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: HR295236
Referred BY	: Self	Sample Type	: Whole blood EDTA
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 06:44 PM.

Test Description	Value(s)	Unit(s)	Reference Range

Erythrocyte Sedimentation Rate (ESR)

ESR - Erythrocyte Sedimentation Rate *MODIFIED WESTERGREN*	5	mm/hr	0 - 10

Interpretation:
ESR is also known as Erythrocyte Sedimentation Rate. An ESR test is used to assess inflammation in the body. Many conditions can cause an abnormal ESR, so an ESR test is typically used with other tests to diagnose and monitor different diseases. An elevated ESR may occur in inflammatory conditions including infection, rheumatoid arthritis ,systemic vasculitis, anemia, multiple myeloma , etc. Low levels are typically seen in congestive heart failure, polycythemia ,sickle cell anemia, hypo fibrinogenemia , etc.

Reference- Dacie and lewis practical hematology

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: HR295236
Referred BY	: Self	Sample Type	: Whole blood EDTA
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 07:52 PM.

Test Description	Value(s)	Unit(s)	Reference Range

HbA1C (Glycosylated Haemoglobin)

Glycosylated Hemoglobin (HbA1c) *HPLC*	4.5	%	< 5.7
Estimated Average Glucose	82.45		

Interpretation:

Interpretation For HbA1c% As per American Diabetes Association (ADA)

Reference Group	HbA1c in %
Non diabetic adults >=18 years	<5.7
At risk (Prediabetes)	5.7 - 6.4
Diagnosing Diabetes	>= 6.5
Therapeutic goals for glycemic control	Age > 19 years Goal of therapy: < 7.0 Age < 19 years Goal of therapy: <7.5

Note:

1. Since HbA1c reflects long term fluctuations in the blood glucose concentration, a diabetic patient who is recently under good control may still have a high concentration of HbA1c. Converse is true for a diabetic previously under good control but now poorly controlled.
2. Target goals of < 7.0 % may be beneficial in patients with short duration of diabetes, long life expectancy and no significant cardiovascular disease. In patients with significant complications of diabetes, limited life expectancy or extensive co-morbid conditions, targeting a goal of < 7.0 % may not be appropriate.

Comments :

HbA1c provides an index of average blood glucose levels over the past 8 - 12 weeks and is a much better indicator of long term glycemic control as compared to blood and urinary glucose determinations ADA criteria for correlation between HbA1c & Mean plasma glucose levels.

HbA1c(%)	Mean Plasma Glucose (mg/dL)	HbA1c(%)	Mean Plasma Glucose (mg/dL)
6	126	12	298
8	183	14	355
10	240	16	413

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423158
Referred BY	: Self	Sample Type	: FLUORIDE F
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:52 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range

Glucose Fasting

Glucose Fasting *Hexokinase*	70	mg/dL	70 - 100

Interpretation:

Status	Fasting plasma glucose in mg/dL
Normal	<100
Impaired fasting glucose	100 - 125
Diabetes	=>126

Reference : American Diabetes Association

Comment :
Blood glucose determinations in commonly used as an aid in the diagnosis and treatment of diabetes. Elevated glucose levels (hyperglycemia) may also occur with pancreatic neoplasm, hyperthyroidism, and adrenal cortical hyper function as well as other disorders. Decreased glucose levels (hypoglycemia) may result from excessive insulin therapy insulinoma, or various liver diseases.

Note
1.The diagnosis of Diabetes requires a fasting plasma glucose of > or = 126 mg/dL or a random / 2 hour plasma glucose value of > or = 200 mg/dL with symptoms of diabetes mellitus.
2.Very high glucose levels (>450 mg/dL in adults) may result in Diabetic Ketoacidosis.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range

Liver Function Test (LFT)

Bilirubin Total *Photometric*	**1.7**	mg/dL	0.2 - 1.2
Bilirubin Direct * *Diazo Reaction*	0.5	mg/dL	0.0 - 0.5
Bilirubin Indirect * *Calculation (T Bil - D Bil)*	**1.2**	mg/dL	0.1 - 1.0
SGOT/AST *IFCC without P5P*	21.2	U/L	5 - 34
SGPT/ALT *IFCC without P5P*	17.9	U/L	0 to 55
SGOT/SGPT Ratio *	1.18	-	-
Alkaline Phosphatase *IFCC*	68	U/L	40 - 150
Total Protein *Biuret*	7.4	g/dL	6.4 - 8.3
Albumin *BCG*	4.7	gm/dL	3.8 - 5.0
Globulin * *Calculation (T.P - Albumin)*	2.7	g/dL	2.3 - 3.5
Albumin :Globulin Ratio * *Calculation (Albumin/Globulin)*	1.74	-	1.0 - 2.1
Gamma Glutamyl Transferase (GGT) * *Photometric*	15	U/L	12 - 64

Interpretation:
The liver filters and processes blood as it circulates through the body. It metabolizes nutrients, detoxifies harmful substances, makes blood clotting proteins, and performs many other vital functions. The cells in the liver contain proteins called enzymes that drive these chemical reactions. When liver cells are damaged or destroyed, the enzymes in the cells leak out into the blood, where they can be measured by blood tests Liver tests check the blood for two main liver enzymes. Aspartate aminotransferase (AST),SGOT: The AST enzyme is also found in muscles and many other tissues besides the liver. Alanine aminotransferase (ALT), SGPT: ALT is almost exclusively found in the liver. If ALT and AST are found together in elevated amounts in the blood, liver damage is most likely present. Alkaline Phosphatase and GGT: Another of the liver's key functions is the production of bile, which helps digest fat. Bile flows through the liver in a system of small tubes (ducts), and is eventually stored in the gallbladder, under the liver. When bile flow is slow or blocked, blood levels of certain liver enzymes rise: Alkaline phosphatase Gamma-utamyl transpeptidase (GGT) Liver tests may check for any or all of these enzymes in the blood. Alkaline phosphatase is by far the most commonly tested of the three. If alkaline phosphatase and GGT are elevated, a problem with bile flow is most likely present. Bile flow problems can be due to a problem in the liver, the gallbladder, or the tubes connecting them. Proteins are important building blocks of all cells and tissues. Proteins are necessary for your body's growth, development, and health. Blood contains two classes of protein, albumin and globulin. Albumin proteins keep fluid from leaking out of blood vessels. Globulin proteins play an important role in your immune system. Low total protein may

Indicate:
1.Bleeding
2.Liver disorder
3.Malnutrition
4.Agammaglobulinemia High Protein levels 'Hyperproteinemia: May be seen in dehydration due to inadequate water intake or to excessive water loss (eg, severe vomiting, diarrhea, Addison's disease and diabetic acidosis) or as a result of increased production of proteins Low albumin levels may be

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range

Caused by:
1.A poor diet (malnutrition).
2.Kidney disease.
3.Liver disease. High albumin levels may be caused by: Severe dehydration.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Kidney Function Test (KFT)

Test Description	Value(s)	Unit(s)	Reference Range
Blood Urea *Urease*	21.2	mg/dL	19 - 44.1
Bun * *Urease*	9.91	mg/dL	8.9 - 20.6
Creatinine *Photometric*	0.74	mg/dL	0.72 - 1.25
eGFR (CKD-EPI)	124.99	ml/min/1.73 sq m	Normal Or High: >= 90 Mild Or Decrease: 60-89 Mild To Moderate Decrease: 45-59 Mild To Severe Decrease: 30-44 Severe Decrease: 15-29 Kidney Failure: < 15
Bun/Creatinine Ratio * *Calculated*	13.39		12 - 20
Urea / Creatinine Ratio * *Calculated*	28.65		25.68- 42.8
Uric Acid *Uricase*	5.9	mg/dL	3.5 - 7.2
Calcium Serum *Arsenazo III*	9.6	mg/dL	8.4 - 10.2
Phosphorus *Photometric*	3	mg/dL	2.3 - 4.7
Sodium *Potentiometric*	140	mmol/L	136 - 145
Potassium *Potentiometric*	4.5	mmol/L	3.5 - 5.1
Chloride *Potentiometric*	105	mmol/L	98 - 107

Interpretation:

Kidney function tests is a collective term for a variety of individual tests and procedureshat can be done toevaluate how well the kidneys are functioning. Many conditions can affect the ability of the kidneys to carryout their vital functions. Somelead to a rapid (acute) decline in kidney functionothers lead to a gradual (chronic) declineinfunction. Both result in a buildup of toxic waste subst done on urine samples, as well as on blood samples. A number of symptoms may indicate a problem with your kidneys. These include : high blood pressure,blood in urine frequent urges to urinate,difficulty beginning urination,painful urination,swelling in the hands and feet due to a buildup of fluids in the body. A single symptom may not mean something serious. However, when occurring simultaneously, these symptoms suggest that your kidneys are not working properly. Kidney function tests can help determine the reason. Electrolytes are present in the human body and the balancing act of the electrolytes in our bodies is essential for normal function of our cells and organs. There has to be a balance.Ionized calcium this test if you have signs of kidney or parathyroid disease. The test may also be done to monitor progress and treatment of these diseases.

"eGFR test is applicable for patients aged 18 years or more."

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Lipid Profile

Test Description	Value(s)	Unit(s)	Reference Range
Total Cholesterol *Enzymatic - Cholesterol Oxidase*	**215**	mg/dL	<200
Triglycerides *Colorimetric - Lip/Glycerol Kinase*	82	mg/dL	<150
HDL Cholesterol *Accelerator Selective Detergent*	41.3	mg/dL	>40
Non HDL Cholesterol * *Calculated*	**173.7**	mg/dL	<130
LDL Cholesterol * *Calculated*	**157.3**	mg/dL	<100
V.L.D.L Cholesterol * *Calculated*	16.4	mg/dL	< 30
Chol/HDL Ratio * *Calculated*	**5.21**	Ratio	3.5 - 5.0
HDL/ LDL Ratio * *Calculated*	**0.26**	Ratio	0.5 - 3.0 Borderline : 3.1 - 6.0 High : > 6.0
LDL/HDL Ratio * *Calculated*	3.81	Ratio	-

Interpretation:
Lipid level assessments must be made following 9 to 12 hours of fasting, otherwise assay results might lead to erroneous interpretation. NCEP recommends of 3 different samples to be drawn at intervals of 1 week for harmonizing biological variables that might be encountered in single assays.

National Lipid Association Recommendations (NLA-2014)	Total Cholesterol (mg/dL)	Triglyceride (mg/dL)	LDL Cholesterol (mg/dL)	Non HDL Cholesterol (mg/dL)
Optimal	<200	<150	<100	<130
Above Optimal			100-129	130 - 159
Borderline High	200-239	150-199	130-159	160 - 189
High	>=240	200-499	160-189	190 - 219
Very High	-	>=500	>=190	>=220

HDL Cholesterol	
Low	High
<40	>=60

Risk Stratification for ASCVD (Atherosclerotic Cardiovascular Disease) by Lipid Association of India.

Risk Category	A. CAD with > 1 feature of high risk group
Extreme risk group	B. CAD with >1 feature of very high risk group of recurrent ACS (within 1 year) despite LDL-C <or = 50 mg/dl or poly vascular disease

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	: **Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

NABL M(EL)T LABS
NABL-M(EL)T-00362

Test Description	Value(s)	Unit(s)	Reference Range

Very High Risk	1.Established ASCVD 2.Diabetes with 2 major risk factors of evidence of end organ damage 3. Familial Homozygous Hypercholesterolemia
High Risk	1. Three major ASCVD risk factors 2. Diabetes with 1 major risk factor or no evidence of end organ damage 3. CHD stage 3B or 4. 4 LDL >190 mg/dl 5. Extreme of a single risk factor 6. Coronary Artery Calcium - CAC > 300 AU 7. Lipoprotein a >/= 50 mg/dl 8. Non stenotic carotid plaque
Moderate Risk	2 major ASCVD risk factors
Low Risk	0-1 major ASCVD risk factors
Major ASCVD (Atherosclerotic cardiovascular disease) Risk Factors	
1. Age >/=45 years in Males & >/= 55 years in Females	3. Current Cigarette smoking or tobacco use
2. Family history of premature ASCVD	4. High blood pressure
5. Low HDL	

Newer treatment goals and statin initiation thresholds based on the risk categories proposed by Lipid Association of India in 2020.

Risk Group	Treatment Goals		Consider Drug Therapy	
	LDL-C (mg/dl)	Non-HDL (mg/dl)	LDL-C (mg/dl)	Non-HDL (mg/dl)
Extreme Risk Group Category A	<50 (Optional goal <OR = 30)	<80 (Optional goal <OR = 60)	>OR = 50	>OR = 80
Extreme Risk Group Category B	>OR = 30	>OR = 60	> 30	> 60
Very High Risk	<50	<80	>OR = 50	>OR = 80
High Risk	<70	<100	>OR = 70	>OR = 100
Moderate Risk	<100	<130	>OR = 100	>OR = 130
Low Risk	<100	<130	>OR = 130*	>OR = 160

*** After an adequate non-pharmacological intervention for at least 3 months.**

References : Management of Dyslipidaemia for the Prevention of Stroke : Clinical practice Recommendations from the Lipid Association of India. Current Vascular Pharmacology,2022,20,134-155.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

High Sensitivity C-Reactive Protein (Hs-CRP)

Test Description	Value(s)	Unit(s)	Reference Range
HIGHLY SENSITIVE C-REACTIVE PROTEIN (hs-CRP) *immunoturbidimetric*	0.8	mg/L	< 1.00

Interpretation:

Cardio CRP In mg/L	Cardiovascular Risk
<1	Low
1-3	Average
3-10	High
>10	Persistent elevation may represent Non cardiovascular inflammation

Note: To assess vascular risk, it is recommended to test hsCRP levels 2 or more weeks apart and calculate the average

Comments:
High sensitivity C Reactive Protein (hsCRP) significantly improves cardiovascular risk assessment as it is a strongest predictor of future coronary events. It reveals the risk of future Myocardial infarction and Stroke among healthy men and women, independent of traditional risk factors. It identifies patients at risk of first Myocardial infarction even with low to moderate lipid levels. The risk of recurrent cardiovascular events also correlates well with hsCRP levels. It is a powerful independent risk determinant in the prediction of incident Diabetes.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 06:30 PM.

Test Description	Value(s)	Unit(s)	Reference Range

Vitamin B12 / Cyanocobalamin

Vitamin - B12 *CMIA*	**153**	pg/mL	187 - 883

Interpretation:
Low Values are a sign of a vitamin B12 deficiency. People with this deficiency are likely to have or develop symptoms.
Causes of vitamin B12 deficiency include:Not enough vitamin B12 in diet (rare except with a strict vegetarian diet), Diseases that cause malabsorption (for example, celiac disease and Crohn's disease), Lack of intrinsic factor, Above normal heat production (for example, with hyperthyroidism), Pregnancy. Increased vitamin B12 levels are uncommon. Usually excess vitamin B12 is removed in the urine. Conditions that can increase B12 levels include: Liver disease (such as cirrhosis or hepatitis), Myeloproliferative disorders (for example, polycythemia vera and chronic myelocytic leukemia).
Vitamin B12: Low Levels can cause malabsorption, Lack of intrinsic factor, Above normal heat production (for example, with hyperthyroidism), Pregnancy.High Level Liver disease, Myeloproliferative disorders (for example, polycythemia vera and chronic myelocytic leukemia).
1. Out of 140 healthy indian population, 91% of Vitamin B 12 concentrations was at lower level: 59.00 pg/ml and upper level: 700.00 pg/ml

"Patients on Biotin supplement may have interference in some immunoassays. Ref: Arch Pathol Lab Med—Vol 141, November 2017. With individuals taking high dose Biotin (more than 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended."

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 06:30 PM.

Test Description	Value(s)	Unit(s)	Reference Range
Vitamin D 25 Hydroxy			
Vitamin D 25 - Hydroxy *CMIA*	**6.0**	ng/mL	Deficiency : <30 ng/mL

Interpretation:
25-Hydroxy vitamin D represents the main body reservoir and transport form. Mild to moderate deficiency is associated with Osteoporosis / Secondary Hyperparathyroidism while severe deficiency causes Rickets in children and Osteomalacia in adults. Prevalence of Vitamin D deficiency is approximately >50% specially in the elderly. This assay is useful for diagnosis of vitamin D deficiency and Hypervitaminosis D. It is also used for differential diagnosis of causes of Rickets & Osteomalacia and for monitoring Vitamin D replacement therapy.

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

Test Description	Value(s)	Unit(s)	Reference Range
Thyroid Profile Total			
Triiodothyronine (T3) *CMIA*	122.2	ng/dL	35 - 193
Total Thyroxine (T4) *CMIA*	7.5	µg/dL	4.87 - 11.2
Thyroid Stimulating Hormone (Ultrasensitive) *CMIA*	1.4	µIU/mL	0.35 - 4.94

Interpretation:

Pregnancy	Reference ranges TSH
1st Trimester	0.1 - 2.5
2nd Trimester	0.2 - 3.0
3rd Trimester	0.3 - 3.0

Note:
TSH levels are subject to circadian variation, reaching peak levels between 2-4 am. and at a minimum between 6-10 pm. The variation is of 50 %, hence time of the day has influence on the measured serum TSH concentrations.

Clinical Use:
- Diagnose Hypothyroidism and Hyperthyroidism
- Monitor T4 replacement or T4 suppressive therapy
- Qunatify TSH levels in the subnormal range

Increased Levels : Primary hypothyroidism, Subclinical hypothyroidis, TSH dependent Hyperthyroidism, Thyroid hormone resistance
Decreased Levels: Grace disease, Autonomous thyroid hormone secretion, TSH deficiency

Primary malfunction of the thyroid gland may result in excessive (hyper) or below normal (hypo) release of T3 or T4. In addition as TSH directly affects thyroid function, malfunction of the pituitary or the hypo - thalamus influences the thyroid gland activity. Disease in any portion of the thyroid-pitutary-hypothala- mus system may influence the levels of T3 and T4 in the blood. In primary hypothyroidism, TSH levels are significantly elevated, while in secondary and tertiary hypothyroidism, TSH levels may be low. In addition, in the Euthyroid Sick Syndrome, multiple alterations in serum thyroid function test findings have been recognized in patients with a wide variety of non-thyroidal illnesses (NTI) without evidence of preexisting thyroid or hypothalami c-pitutary diseases. Thyroid Binding Globulin (TBG) concentrations remain relatively constant in healthy individuals. However, pregnancy, excess estrogen's, androgen's, antibiotic steroids and glucocorticoids are known to alter TBG levels and may cause false thyroid values for Total T3 and T4 tests.

TSH	T4	T3	INTERPRETATION
High	Normal	Normal	Mild (subclinical) hypothyroidism
High	Low	Low or Normal	Hypothyroidism
Low	Normal	Normal	Mild (subclinical) hyperthyroidism
Low	High or normal	High or normal	Hyperthyroidism
Low	Low or normal	Low or normal	Nonthyroidal illness; pituitary (secondary) hypothyroidism

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

Test Description	Value(s)	Unit(s)	Reference Range
Normal High High Thyroid hormone resistance syndrome (a mutation in the thyroid hormone receptor decreases thyroid hormone function)			

Estimated Glomerular Filtration Rate (eGFR)

Creatinine *Photometric*	0.74	mg/dL	0.72 - 1.25
eGFR (CKD-EPI)	124.99	ml/min/1.73 sq m	Normal Or High: >= 90 Mild Or Decrease: 60-89 Mild To Moderate Decrease: 45-59 Mild To Severe Decrease: 30-44 Severe Decrease: 15-29 Kidney Failure: < 15

Interpretation:

1. The CKD-EPI equation, expressed as a single equation, is:

$GFR = 141 * \min(Scr/\kappa,1)^{\alpha} * \max(Scr/\kappa, 1)^{-1.209} * 0.993^{Age} * 1.018$ [if female] * 1.159 [if black]

Scr is serum creatinine (mg/dL), κ is 0.7 for females and 0.9 for males, α is -0.329 for females and -0.411 for males, min indicates the minimum of Scr/κ or 1, and max indicates the maximum of Scr/κ or 1.

2. The CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) equation was developed in an effort to create a more precise formula to estimate glomerular filtration rate (GFR) from serum creatinine and other readily available clinical parameters, especially at when actual GFR is >60 mL/min per $1.73m^2$.

Reference:

Levey et al. Annals of Internal Medicine 2009 May 5, 150 (9): 604-12

"eGFR test is applicable for patients aged 18 years or more."

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

Patient NAME	**: Mr Sourabh Jain**		
DOB/Age/Gender	: 30 Y/Male	Report STATUS	: Final Report
Patient ID / UHID	: 11363161/RCL10544339	Barcode NO	: ZH423157
Referred BY	: Self	Sample Type	: Serum
Sample Collected	: Feb 12, 2025, 08:52 AM	Report Date	: Feb 12, 2025, 05:55 PM.

Test Description	Value(s)	Unit(s)	Reference Range

Iron

Iron *Ferene*	110	µg/dL	65 - 175

Interpretation:
Iron is body's essential trace element used for differential diagnosis of anemias, diagnosis of hemochromatosis and hemosiderosis.

Apolipoproteins A1 & B

Apolipoprotein A-1 (APO-A) *Tina-quant*	134		104 - 202
Apolipoprotein B (APO-B)	107		
Apo B / Apo A1 Ratio	0.8		

Interpretation:
The Apo A1 and Apo B blood tests are crucial for assessing lipid metabolism and cardiovascular risk. High Apo A1 and low Apo B levels are associated with a lower risk of cardiovascular disease, while low Apo A1 and high Apo B levels indicate a higher risk. Management involves lifestyle modifications, medications, and regular monitoring to maintain optimal lipid levels and reduce cardiovascular risk. Consulting healthcare providers for accurate interpretation and tailored treatment plans is essential for effective management.

*** End Of Report ***

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

Dr. Arvind Kumar
MD (Pathology)
Consultant Pathologist

HOW WAS YOUR EXPERIENCE?

We value your feedback, please scan with your phone camera
and rate overall experience.

How likely are you to recommend Redcliffe Labs to your family & friends

1	2	3	4	5	6	7	8	9	10

Not Likely — Very Likely

Scan with your phone camera